

Excel Project Report: Lead Conversion & Marketing Campaign Analysis

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Date : May 28, 2026

Abstract

This report presents a comprehensive analysis of Knovist Technologies' lead conversion and marketing campaign dataset using Microsoft Excel. The project focused on cleaning and preprocessing the dataset, handling missing values, identifying duplicate records and outliers, and analyzing campaign performance through pivot tables, charts, and dashboards.

The dataset contained 27,375 records and 20 columns related to customer leads, campaign performance, sales activity, and marketing costs. Various Excel techniques such as Remove Duplicates, Conditional Formatting, QUARTILE, IQR calculations, Pivot Tables, Charts, and Dashboard Visualizations were used throughout the project.

The project successfully improved dataset quality and generated business insights related to campaign effectiveness, employee performance, lead conversion, and marketing costs.

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1. Executive Summary

This project focused on cleaning and analyzing the lead conversion and marketing campaign dataset of Knovist Technologies using Microsoft Excel.

The project involved:

- handling missing values
- checking duplicate records
- detecting outliers
- validating emails and contact numbers
- correcting formatting issues
- performing descriptive analysis
- creating pivot tables and dashboards

Key findings include:

- approximately 24.5% of leads had no first call attempt
- the references column was completely empty and removed
- no duplicate records were found
- some total_cost values behaved as outliers
- social media campaigns generated a large number of leads

The dataset quality improved significantly after preprocessing and became suitable for reporting and business analysis.

2. Introduction & Objectives

2.1 Objective

The objective of this project was to analyze lead conversion and marketing campaign data to improve data quality and generate business insights using Microsoft Excel.

The project aimed to:

- clean missing and inconsistent data
- identify duplicate records
- detect abnormal values
- analyze lead conversion performance
- improve reporting accuracy
- create interactive visualizations and dashboards

2.2 Key Business Questions

1. Are there missing or inconsistent values in the dataset?
2. Are duplicate records present in the dataset?
3. Which columns contain outliers or abnormal values?
4. Which marketing campaigns generated better sales?
5. Which source platforms produced more leads and conversions?
6. How effective were employees in converting leads?
7. How can dashboards improve business reporting and decision-making?

3. Data & Methodology

3.1 Data Source

The dataset used for this project was : Live Project Dataset.csv

The dataset contained : 27,375 rows and 20 columns

related to:

- customer leads
- sales activity
- marketing campaigns
- conversion performance
- campaign costs

3.2 Data Preprocessing & Cleaning

The dataset initially contained:

- missing values
- formatting inconsistencies
- empty columns
- abnormal numerical values

The following preprocessing steps were performed:

Missing Value Handling

Blank values were replaced with meaningful labels such as:

- “Not Attempted”

- “No Tag”
- “NaN”
- “NaN”

The:

References

column was completely empty and removed from the dataset

Data Formatting

Formatting corrections included:

- renaming column headers
- removing spaces
- converting column names to lowercase
- formatting dates properly
- formatting numerical columns

Validation

Customer emails and contact numbers were validated to identify invalid entries.

3.3 Analytical Methods

The following Excel techniques were used:

- Remove Duplicates
- Conditional Formatting
- COUNTIF formula
- QUARTILE and IQR calculations
- Pivot Tables
- Charts and Dashboard Visualization
- Descriptive Statistics

4. Data Preprocessing & Cleaning

4.1 Missing Value Analysis

The dataset contained missing values in several columns. The `first_call_attempt`, `last_call_date`, and `last_call_time` columns each contained 6,729 missing values, indicating that many leads did not have recorded call activities. The `lead_tag` column contained 11,416 missing values, while the `references` column contained 27,375 missing values and was completely empty throughout the dataset.

4.2 Missing Value Treatment

The following actions were performed:

The missing values in the `first_call_attempt` column were filled with "Not Attempted", while the missing values in the `lead_tag` column were replaced with "No Tag". Similarly, the `last_call_date` column was filled with "NaN", and the `last_call_time` column was filled with "NaN". The `references` column was completely empty throughout the dataset, so it was removed during the data cleaning process.

4.3 Column Name Cleaning

Column names were standardized for consistency.

Column names were cleaned and standardized by converting them into lowercase format and replacing spaces with underscores. For example, `Total Leads` was renamed as `total_leads`, `Conversion Rate (%)` was changed to `conversion_rate`, and `Total Cost` was renamed as `total_cost` to improve consistency and readability in the dataset.

4.4 Data Formatting

- Date columns were converted into short date format.
- Numeric columns were formatted as numbers.
- Extra spaces were removed using `TRIM`.
- Text formatting was corrected using `PROPER` function.

5. Duplicate Record Analysis

Duplicate records were checked using:

- Remove Duplicates tool
- Conditional Formatting
- `COUNTIF` formula

Result

No duplicate records were found in the dataset.

This indicates that the dataset maintained good uniqueness and consistency.

6. Outlier Detection

Outlier analysis was performed on numerical columns such as:

- total_leads
- sales
- cost_per_lead
- total_cost

6.1 Conditional Formatting Method

Conditional Formatting was used to highlight unusually high values in the:

total_cost column.

6.2 IQR Method

The Interquartile Range (IQR) method was used to identify abnormal values.

Formulas Used

Q1 =QUARTILE(T:T,1)

Q3 =QUARTILE(T:T,3)

IQR =Q3-Q1

Lower Limit =Q1-(1.5*IQR)

Upper Limit =Q3+(1.5*IQR)

Values outside the limits were considered outliers.

7. Descriptive Statistics

Descriptive statistics were calculated for important numerical columns.

Metrics Used

- Average
- Median
- Standard Deviation

Numerical Features Analyzed

- total_leads
- sales
- conversion_rate
- total_cost

The descriptive analysis helped understand data distribution, variation, and business performance patterns.

8. Pivot Table Analysis & Insights

8.1 Campaign Performance Analysis

A pivot table was created using:

- Rows → campaign_name
- Values → sales

Findings

Some marketing campaigns generated significantly higher sales compared to others.

Visualization Used

- Bar Chart

Business Insight

High-performing campaigns should receive higher marketing investment and strategic focus.

8.2 Lead Source Analysis

A pivot table was created using:

- Rows → source_type

- Values → sales

Findings

Social media platforms generated a large number of successful leads and sales.

Visualization Used

- Pie Chart

Business Insight

High-performing platforms can improve customer acquisition efficiency.

8.3 Employee Performance Analysis

A pivot table was created using:

- Rows → assigned_to
- Values → sales

Findings

Some employees handled and converted leads more effectively.

Visualization Used

- Column Chart

Business Insight

Employee performance analysis helps improve training and productivity tracking.

8.4 Lead Status Distribution

A pivot table was created using:

- Rows → lead_status
- Values → count of lead_status

Findings

Many leads remained in follow-up stages while some were successfully converted.

Visualization Used

- Donut Chart

Business Insight

Monitoring lead stages helps improve sales pipeline management.

9. Dashboard Development

An interactive dashboard was created in Microsoft Excel to visualize business performance and marketing analytics.

9.1 Dashboard Components

KPI Cards

The dashboard included:

- Total Leads
- Total Sales
- Average Conversion Rate
- Total Cost

Charts Used

A bar chart was used to analyze sales by campaign, a pie chart showed source type distribution, a column chart visualized employee performance, and a donut chart represented lead status distribution.

9.2 Interactive Features

The dashboard included slicers for:

- campaign_name
- source_type
- lead_status
- assigned_to

The slicers enabled interactive filtering of charts and pivot tables.

9.3 Dashboard Benefits

The dashboard helped visualize:

- campaign effectiveness
- sales trends
- employee performance
- lead conversion metrics
- marketing costs

The dashboard improved reporting quality and business decision-making.

10. Conclusion

This project successfully demonstrated the process of cleaning, preprocessing, and analyzing a real-world lead conversion and marketing campaign dataset using Microsoft Excel.

The dataset initially contained missing values, formatting inconsistencies, empty columns, and abnormal numerical values. Data cleaning techniques such as handling missing values, removing empty columns, correcting formatting issues, validating customer information, and detecting outliers significantly improved data quality.

Further analysis using descriptive statistics, pivot tables, charts, and dashboards generated meaningful business insights related to campaign performance, sales conversion, employee productivity, and marketing effectiveness.

Overall, the cleaned dataset and interactive dashboard provided a powerful analytical solution for improving business reporting and supporting data-driven decision-making at Knovista Technologies.